Trend Following: Bollinger Bands + EMA Filter

BTC/USD Spot | Coinbase | 60-Min Bars | Long-Only | Jan 2023 - Nov 2025

1. Market & Instrument

Exchange	Instrument	Mode	Direction	Timeframe
Coinbase (Advanced)	BTC/USD Spot	Spot (1x, no leverage)	Long-only	60 min

2. Trading Hypothesis

This strategy exploits momentum persistence in Bitcoin. When BTC breaks above the upper Bollinger Band while remaining above a long-term EMA, it signals the onset of a strong directional move. Crypto markets exhibit pronounced trending regimes driven by herding, FOMO-driven inflows, and reflexive price-liquidity feedback loops. The combination of a volatility breakout (BB) with a slow trend filter (EMA) avoids mean-reversion traps and only enters when both short-term and long-term momentum align. The strategy is naturally selective (67 trades over 2.9 years), waiting for high-conviction setups only.

3. Strategy Rules

Indicators & Parameter Rationale

Indicator	Parameter	Rationale
EMA	1000 bars (~42 days)	Captures multi-week trend; filters noise from daily volatility
BB Window	75 bars (~3 days)	Balances responsiveness vs false breakout frequency
BB Std Dev	3.0	High threshold = only enters on extreme breakouts (high conviction)

Parameters validated via Optuna (150 trials): top 10 trials all converged to ema=900-1000, bb_window=75, bb_dev=3.0, confirming these are in a robust optimum region, not an isolated peak.

Entry & Exit

	Condition
ENTRY (Long)	Close > BB Upper(75, 3.0) AND Close > EMA(1000)
EXIT	Close < BB Middle(75-bar SMA) OR Close < EMA(1000)

Signal Logic (Pseudocode)

```
ema      = EMA(close, 1000)
bb_upper = SMA(close, 75) + 3.0 * StdDev(close, 75)
bb_mid   = SMA(close, 75)
if close > bb_upper AND close > ema:  signal = LONG
if close < bb_mid   OR  close < ema:  signal = EXIT
signal is forward-filled (hold position until exit triggers)
```

Position Sizing & Risk Management

Rule	Setting	Note
Position Size	100% equity	Max 1 concurrent position, no pyramiding
Stop-Loss	15% from entry	Hard safety net; never triggered (worst = -5.7%)
Take-Profit	No cap	Exit via signal only (best trade = +23.4%)
Leverage	1x (spot only)	No margin or futures exposure
Max Risk / Trade	15% of equity	Bounded by stop-loss level

4. Backtest Assumptions

Parameter	Value
Data Source	Coinbase BTC/USD spot OHLCV candles
Period	Jan 1, 2023 - Nov 16, 2025 (2.9 years, 25,220 bars)
Trading Fees	0.16% round-trip (Coinbase Advanced taker fee)
Slippage	0 assumed (hourly bars, BTC/USD highly liquid)
Execution	Filled at bar close price; no partial fills
Starting Capital	$100

5. Results & Evaluation

Key Performance Metrics

Metric	Value	Metric	Value
Total Return	+160.0%	Win Rate	44.8% (30W / 37L)
Ann. Return	+39.4%	Profit Factor	2.36
Volatility	22.2%	Avg Win / Avg Loss	2.92x
Sharpe Ratio	1.78	Avg Trade Return	+1.57%
Max Drawdown	-12.3%	Avg Trade Duration	2.6 days
Return / DD	3.20x	Total Trades	67
Fee Drag	$40.61 (20.2%)	Gross PnL	$200.59

Annual Breakdown

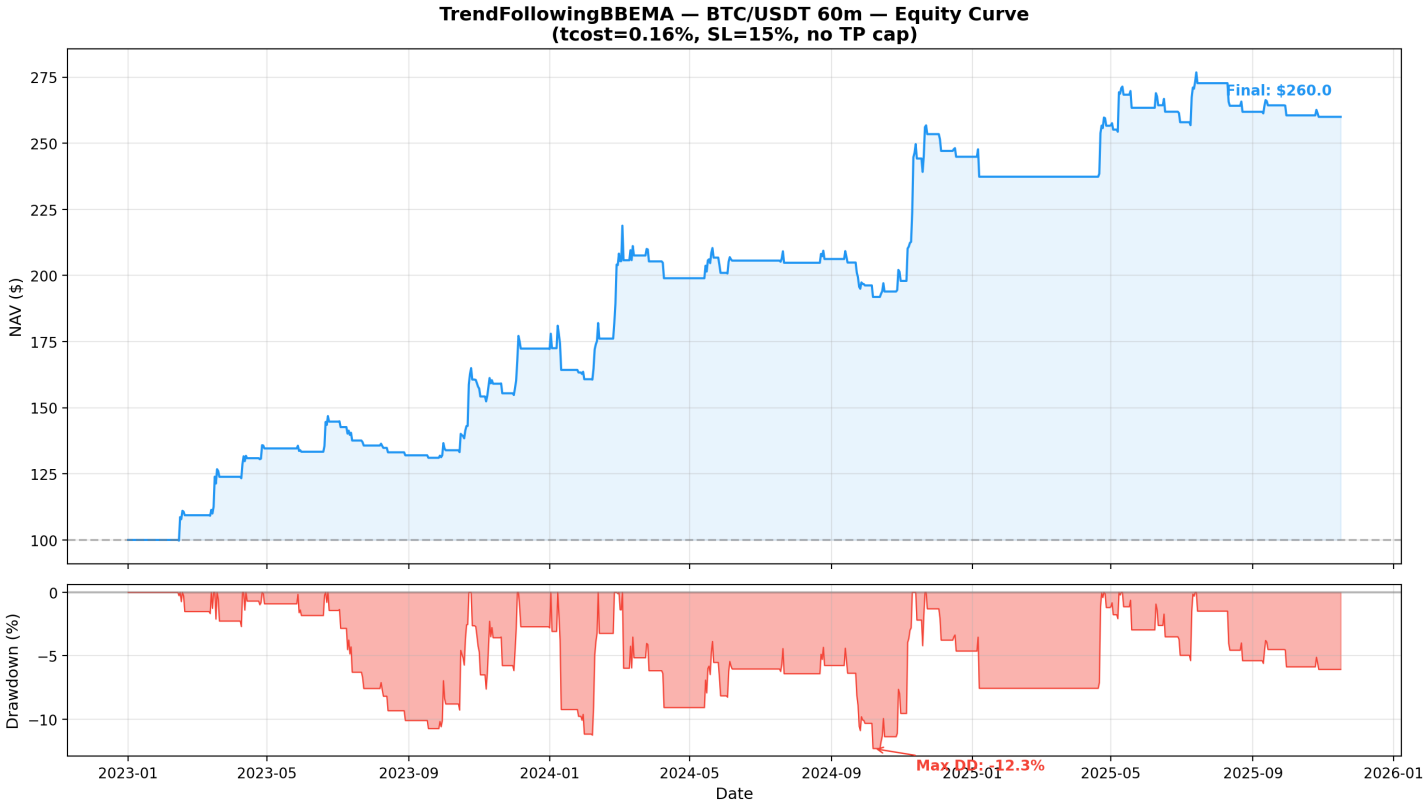
Year	Trades	Net PnL	Win Rate	Return
2023	25	$72.36	40%	+72.4%
2024	28	$72.53	54%	+42.1%
2025 (YTD Nov)	14	$15.09	36%	+6.2%

Trade Return Distribution

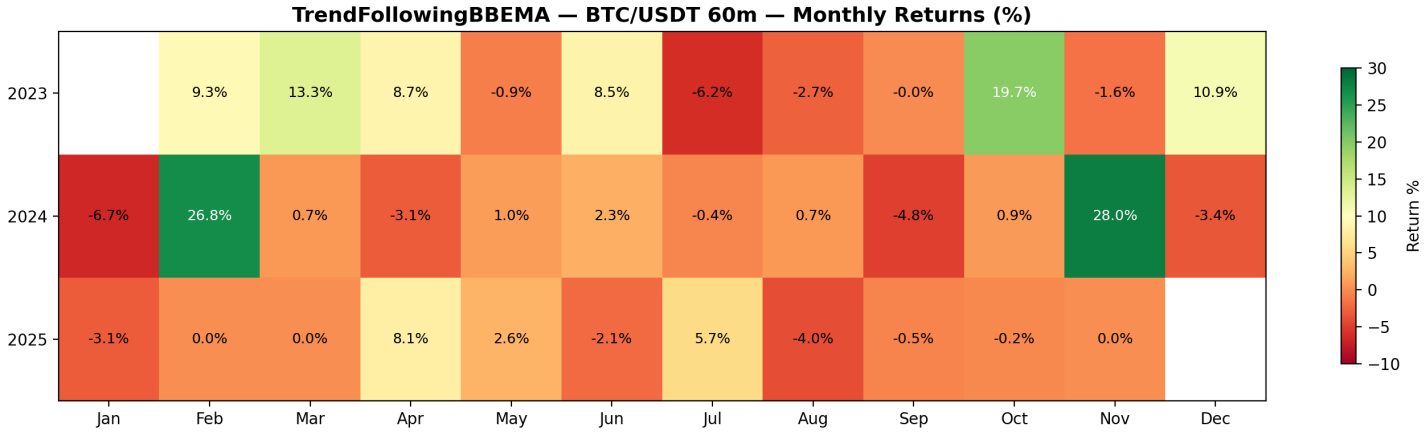
	Min	5th	25th	Median	75th	95th	Max
Return	-5.71%	-2.99%	-1.51%	-0.48%	+2.57%	+14.50%	+23.38%

Positive skew: median slightly negative but fat right tail (top 5 trades avg +16.7%) drives profitability. Characteristic of trend-following: cut losers fast, let winners run.

Equity Curve & Drawdown



Monthly Returns (%)



Strategy vs BTC Buy & Hold

	Strategy	BTC Buy & Hold
Total Return	+160.0%	+468.5%
Ann. Return	+39.4%	+83.0%
Volatility	22.2%	46.2%
Sharpe Ratio	1.78	1.80
Max Drawdown	-12.3%	-27.7%
Time in Market	16.3%	100%

B&H outperforms on absolute return due to the strong BTC bull cycle (2023-2025). However, the strategy achieves a comparable Sharpe (1.78 vs 1.80) with less than half the drawdown (-12% vs -28%) and only 16% time exposure. Capital is free 84% of the time for other opportunities.

Top 5 Winning & Losing Trades

#	Entry Date	Entry $	Exit $	Return	Duration
W1	2024-11-05	$71,134	$88,019	+23.4%	8.6 days
W2	2024-02-26	$52,851	$61,825	+16.6%	5.0 days
W3	2023-03-13	$23,628	$27,608	+16.5%	7.7 days
W4	2023-10-19	$29,010	$33,767	+16.1%	6.6 days
W5	2023-12-01	$38,820	$43,174	+10.9%	6.3 days
L1	2024-01-11	$48,695	$46,068	-5.7%	2 hours
L2	2025-08-10	$121,833	$118,404	-3.1%	1.3 days
L3	2024-04-08	$71,988	$69,985	-3.1%	1.2 days

Exposure & Regime Performance

Regime	Period	Return	Stat	Value
Bull	Sep 23 - Mar 24	+57.7%	Time in market	16.3%
Bull	Sep 24 - Dec 24	+19.8%	Time flat	83.7%
Bull	Apr 25 - Oct 25	+9.8%	Max win streak	4
Bear	Jun - Aug 2024	-0.2%	Max loss streak	9
Bear	Jan - Mar 2025	0.0%	DD recovery	34 days
Sideway	Feb - Sep 2023	+32.0%	Expectancy	$2.39/trade

6. Robustness

Parameter Stability (Optuna) & Walk-Forward OOS

Optuna (150 trials, max Sharpe): best params ema=950, bb=75, dev=3.0, nearly identical to baseline (1000/75/3.0). Top 10 trials all cluster ema=900-1000. Not overfit.

Walk-forward (Jan 2024 - Nov 2025): monthly re-optimization (80 trials), test on next month. Params stable (ema=900-1450, bb=75-95, dev=2.75-3.75).

	Full Backtest (IS)	Walk-Forward (OOS)
Total Return	+160.0%	+17.5%
Sharpe	1.78	0.52
Max DD	-12.3%	-15.1%

IS-to-OOS decay (1.78 -> 0.52) is typical. Strategy remains profitable OOS, confirming edge is not purely in-sample. OOS params converge to same region as full-sample.

7. Critique & Limitations

When It Works / Struggles

- Works: bull markets, post-consolidation breakouts, sideways regimes with intermittent trends.
- Struggles: choppy/whipsaw markets (frequent BB touches without follow-through), sharp V-reversals (exit signals lag), prolonged bear markets (capital sits idle).
- 2025 degradation (+6.2% YTD) suggests the edge may be weakening in the current regime.

Key Risks

- Regime change: if BTC becomes structurally mean-reverting, trend logic generates persistent losses.
- Fee sensitivity: fees consume 20.2% of gross PnL. At 0.25% round-trip, strategy becomes marginal.
- Small sample: 67 trades over 2.9 years. Profitability driven by ~5 large winners (top 5 = majority of PnL). Removing best 2-3 trades would substantially reduce returns.
- Concentration risk: 100% equity in single BTC position. Black swan events directly impact portfolio. 15% SL provides partial protection.
- WF OOS Sharpe (0.52) well below IS (1.78), suggesting part of the backtest edge may not persist.

Tools: Python (pandas, ta, optuna, numba). Code and data available upon request.